

Ghost Kohnert posets

Kelsey Hanser* and Nicholas Mayers*

*Department of Mathematics, North Carolina State University, Raleigh, NC, 27605

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Abstract

Recently, Pan and Yu showed that Lascoux polynomials can be defined in terms of certain collections of diagrams consisting of unit cells arranged in the first quadrant. Starting from certain initial diagrams, one forms a finite set of diagrams by applying two types of moves: Kohnert and ghost moves. Both moves cause at most one cell to move to a lower row with ghost moves leaving a new “ghost cell” in its place. Each diagram formed in this way defines a monomial in the associated Lascoux polynomial. Restricting attention to diagrams formed by applying sequences of only Kohnert moves in the definition of Lascoux polynomials, one obtains the family of key polynomials. Recent articles have considered a poset structure on the collections of diagrams formed when one uses only Kohnert moves. In general, these posets are not “well-behaved,” not usually having desirable poset properties. Here, as an intermediate step to studying the analogous posets associated with Lascoux polynomials, we consider the posets formed by restricting attention to those diagrams formed by using only ghost moves. Unlike in the case of Kohnert posets, we show that such “ghost Kohnert posets” are always ranked join semi-lattices. In addition, we establish a necessary condition for when ghost Kohnert posets are bounded and, consequently, lattices.

1 Introduction

In recent work [2, 10, 13, 18, 19], it has been found that many families of polynomials arising in algebraic combinatorics can be viewed as generating polynomials for certain collections of “diagrams.” Such families include Schubert [12], key [6, 7], and Lascoux polynomials [11]. Within this framework, diagrams are collections of unit cells (some of which may be decorated) arranged in the first quadrant (see Figures 1 and 2) which are encoded by (signed) monomials where the power of x_i is the number of cells in row i . For the polynomials known to have such definitions currently, the encoded diagrams are formed by starting from an initial “seed” diagram and applying sequences of appropriately defined moves, i.e., Kohnert or K-Kohnert moves (see below as well as Section 2.1). Such moves effect the position of a single cell in the diagram and can additionally add decorated cells to the diagram as well. Here, we study a family of posets on decorated diagrams which arise within this framework.

Definitions for families of polynomials in terms of diagrams and moves, as described above, originated in the thesis of Kohnert [10] where he showed that key polynomials admit such a definition and conjectured the same for Schubert polynomials; this conjecture was eventually proven by multiple authors [2, 18, 19]. The families of key and Schubert polynomials both form bases of the polynomial ring $\mathbb{Z}[x_1, x_2, \dots]$ which consist of homogeneous polynomials that are indexed by weak compositions in the case of keys and permutations in the case of Schuberts. Given a weak composition $\alpha \in \mathbb{Z}_{\geq 0}^n$ and a permutation $w \in S_n$, we denote the associated key and Schubert polynomials by $\mathcal{K}_\alpha$ and $\mathfrak{S}_w$, respectively. Moreover, we associate with $\mathcal{K}_\alpha$ a corresponding “key diagram,” denoted $\mathbb{D}(\alpha)$, and with $\mathfrak{S}_w$ a corresponding “Rothe diagram,” denoted $\mathbb{D}(w)$. See Figure 1 (a) for $\mathbb{D}(\alpha)$ with the weak composition $\alpha = (1, 1, 3, 2)$ and Figure 1 (d) for $\mathbb{D}(w)$ with the permutation $w = [4, 2, 5, 3, 1]$. For both key and Schubert polynomials, one starts from the associated